Binary Search Fundamentals

Binary Search is a divide-and-conquer search algorithm with logarithmic time complexity O(log n).
Fundamental Precondition: The input array or sequence MUST BE SORTED in order for binary search to work.
How it works: Compare the target value to the middle element of the sorted collection.
If the target matches the middle element, the position is returned immediately.
If the target is smaller than the middle element, search continues in the lower half of the array.
If the target is larger than the middle element, search continues in the upper half of the array.
Why sorting is required: If elements are unsorted, you cannot eliminate half of the remaining elements, because an element could be anywhere in the unsorted portions.
Common edge cases: Empty arrays, duplicate keys, integer overflow when computing mid = (low + high) / 2, which should safely be computed as mid = low + (high - low) // 2.